ISOM 3540 – Introduction to Probability Models

Assignment 4: Solution to Question 5

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Model and notation. Let X be the number of interested customers and Y the number who purchase a policy. Under the assumptions in Question 5,

$$X \sim \text{Bin}(100, 0.5), \qquad Y \mid (X = x) \sim \text{Bin}(x, 0.3).$$

Write $A = \{12 \le Y \le 18\}$, and define $g(x) = P(A \mid X = x)$.

(a) Express the probability as an expectation of a function of X.

For each $x \in \{0, \dots, 100\}$, the conditional binomial model gives

$$g(x) = \begin{cases} 0, & 0 \le x < 12, \\ \displaystyle\sum_{y=12}^{\min\{18,x\}} \binom{x}{y} (0.3)^y (0.7)^{x-y}, & 12 \le x \le 100. \end{cases}$$

By the law of total probability (or the tower property),

$$\boxed{P(12 \le Y \le 18) = E[g(X)] = \sum_{x=0}^{100} \binom{100}{x} (0.5)^{100} g(x).} \tag{1}$$

This is the requested expectation representation; it will also be the basis of the simulation.

(b) Monte Carlo estimate using part (a).

Use $M = 40{,}000$ independent replications and the fixed random seed `35402026`. The procedure is:

1. Generate $X_1, \dots, X_M$ independently from $\text{Bin}(100, 0.5)$.
2. For each realization $X_i = x$, evaluate $g(x)$ using the conditional binomial sum in (a).
3. Average the conditional probabilities rather than generating a second-stage Y_i:

$$\widehat{p}_M = \frac{1}{M} \sum_{i=1}^{M} g(X_i) \quad \Longrightarrow \quad \boxed{\widehat{p}_{40{,}000} = 0.67329525.}$$

The sample standard error of this Monte Carlo mean is $s_g/\sqrt{M} = 0.00030768$, yielding the approximate 95% simulation interval $[0.67269219,\ 0.67389830]$. Because it averages conditional expectations, this estimator avoids the additional sampling variability from explicitly simulating $Y \mid X$.

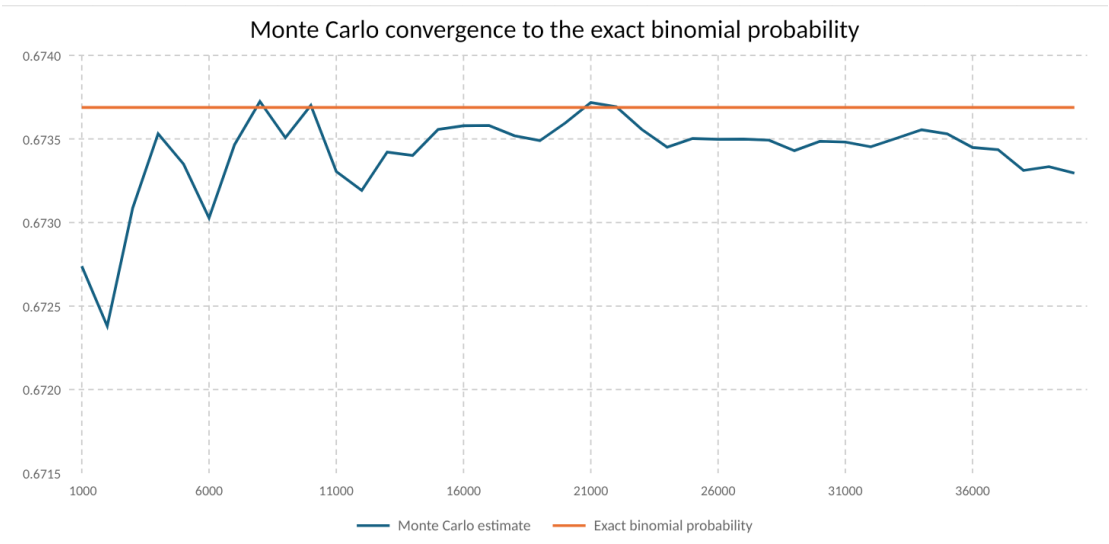


Figure 1: Monte Carlo running estimate versus the exact binomial benchmark; chart created in Excel. The graph is a reproducibility aid, not an additional requirement of Question 5.

(c) Derive the marginal distribution of Y and find the exact probability.

For $y = 0, \ldots, n$, the law of total probability gives

$$\begin{aligned} P(Y = y) &= \sum_{x=y}^{n} P(Y = y \mid X = x) P(X = x) \\ &= \sum_{x=y}^{n} \binom{x}{y} p_2^y (1-p_2)^{x-y} \binom{n}{x} p_1^x (1-p_1)^{n-x}. \end{aligned}$$

Using $\binom{n}{x}\binom{x}{y} = \binom{n}{y}\binom{n-y}{x-y}$ and setting $k = x - y$,

$$\begin{aligned} P(Y = y) &= \binom{n}{y} (p_1 p_2)^y \sum_{k=0}^{n-y} \binom{n-y}{k} [p_1(1-p_2)]^k (1-p_1)^{n-y-k} \\ &= \binom{n}{y} (p_1 p_2)^y [(1-p_1) + p_1(1-p_2)]^{n-y} \\ &= \binom{n}{y} (p_1 p_2)^y (1 - p_1 p_2)^{n-y}. \end{aligned}$$

Therefore $\boxed{Y \sim \mathrm{Bin}(n, p_1 p_2)}$. Here $n = 100$ and $p_1 p_2 = 0.15$, so

$$\boxed{\begin{aligned} P(12 \le Y \le 18) &= \sum_{y=12}^{18} \binom{100}{y} (0.15)^y (0.85)^{100-y} \\ &= 0.6736884056 \;\approx\; 0.67368841. \end{aligned}} \tag{2}$$

The Monte Carlo difference is $\widehat{p}_{40{,}000} - P(A) = -0.00039316$, whose magnitude is about 1.28 estimated Monte Carlo standard errors. The two results are consistent.

(d) Conditional probability that $40 \le X \le 60$, given $12 \le Y \le 18$.

Let $B = \{40 \le X \le 60\}$. By Bayes' rule and the definition of $g(x)$,

$$\begin{aligned} P(B \mid A) = \frac{P(A \cap B)}{P(A)} &= \frac{\displaystyle\sum_{x=40}^{60} P(X = x) P(A \mid X = x)}{P(A)} \\ &= \frac{\displaystyle\sum_{x=40}^{60} \binom{100}{x} (0.5)^{100} \sum_{y=12}^{18} \binom{x}{y} (0.3)^y (0.7)^{x-y}}{\displaystyle\sum_{y=12}^{18} \binom{100}{y} (0.15)^y (0.85)^{100-y}}. \end{aligned}$$

Numerical evaluation yields $P(A \cap B) = 0.6574724968$ and $P(A) = 0.6736884056$. Thus

$$\boxed{P(40 \le X \le 60 \mid 12 \le Y \le 18) = \frac{0.6574724968}{0.6736884056} = 0.9759296603 \;\approx\; \mathbf{0.97592966}.} \tag{3}$$

Independent check. A customer is an interested non-buyer with probability $0.5(0.7) = 0.35$, a buyer with probability $0.5(0.3) = 0.15$, and otherwise uninterested. Conditional on $Y = y$,

$$X \mid (Y = y) \;\overset{d}{=}\; y + \mathrm{Bin}\left(100 - y, \frac{0.35}{0.85}\right).$$

Summing this conditional distribution over $y = 12, \ldots, 18$, weighted by $P(Y = y \mid A)$, reproduces 0.9759296603, confirming the direct joint-probability calculation.

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The Excel workbook contains the 40,000 seeded simulation results, exact conditional-probability lookup, formulas and an editable convergence plot. The linked data and script enable verification of the reported results. All material for assessment is contained in this PDF.